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Differential Accumulation and Elimination Behavior of Perfluoroalkyl Acid Isomers in Occupational Workers in a Manufactory in China

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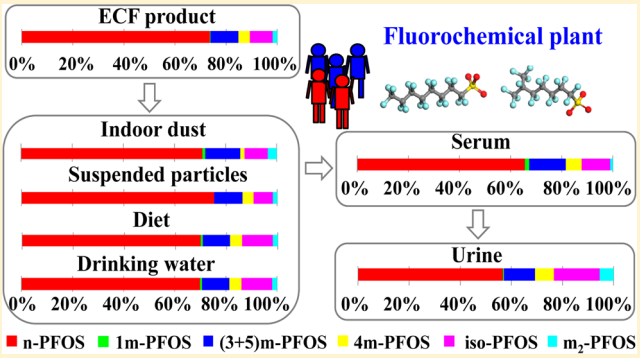
Supporting Information

ABSTRACT: In this study, serum and urine samples were collected from 36 occupational workers in a fluorochemical manufacturing plant in China from 2008 to 2012 to evaluate the body burden and possible elimination of linear and branched perfluoroalkyl acids (PFAAs). Indoor dust, total suspended particles (TSP), diet, and drinking water samples were also collected to trace the occupational exposure pathway to PFAA isomers. The geometric mean concentrations of perfluorooctanesulfonate (PFOS), perfluorooctanoate (PFOA), and perfluorohexanesulfonate (PFHxS) isomers in the serum were 1386, 371, and 863 ng mL^{-1}, respectively. The linear isomer of PFOS, PFOA, and PFHxS was the most predominant PFAA in the serum, with mean proportions of 63.3, 91.1, and 92.7% respectively, which were higher than the proportions in urine. The most important exposure routes to PFAA isomers in the occupational workers were considered to be the intake of indoor dust and TSP. A renal clearance estimation indicated that branched PFAA isomers had a higher renal clearance rate than did the corresponding linear isomers. Molecular docking modeling implied that linear PFOS (*n*-PFOS) had a stronger interaction with human serum albumin (HSA) than branched isomers did, which could decrease the proportion of *n*-PFOS in the blood of humans via the transport of HSA.



INTRODUCTION

Per- and polyfluoroalkyl substances (PFASs) have been widely used in products such as lubricants, textile coatings and fire-fighting foams because they have excellent surfactant properties and thermal stability and are both hydro- and oleophobic.[1] However, perfluoroalkyl acids (PFAAs) such as perfluorooctanesulfonate (PFOS) and perfluorooctanoic acid (PFOA) have recently received much attention because of their persistence, wide distribution in the environment, and potential toxicity.[1]

The production of PFASs began approximately 60 years ago.[2,3] The main manufacturing processes for PFAS-related products include electrochemical fluorination (ECF) and telomerization.[4] The 3M company, formerly the largest producer that used ECF, ceased production of perfluorooctane sulfonyl fluoride (PFOSF) in 2002, but the ECF process is still used in some Asian countries, including China.[2,3,5] Recent studies showed that the ECF process was used to produce approximately 70% of linear PFASs and 30% of the branched isomers, whereas the telomerization process was mostly used to produce linear PFASs.[6,7]

PFAA isomers have been ubiquitously found in the environment, in the biota, and even in humans,[7−13] and they show different properties in organisms, such as different half-lives and bioaccumulation factors.[6] Previous studies on the basis of animal models indicated that linear and branched PFAA isomers have different elimination properties and toxicity.[14−16] However, PFOS isomers in human serum samples showed excretion properties different from those in test animals, which deserves further investigation.[5,17]

Potential exposure pathways to PFASs for the general population include diet, drinking water, indoor dust, and indoor/outdoor air.[18−25] Previous studies noted that workers in fluorochemical production plants are a subgroup that have an exceptionally high body burden of PFAAs.[26,27] However,

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possible sources and routes of exposure to PFAA isomers in occupationally exposed workers are not well-characterized.

Our previous study revealed that high levels of PFAAs were found in the ambient environment of a perfluorosulfonate (PFSA) manufacturing facility.[28] In the present study, we continued to evaluate the levels of linear and branched PFOS, PFOA, and PFHxS isomers in serum and urine samples collected from workers from 2008 to 2012 in the manufacturing facility. Indoor dust and total suspended particles (TSP) from the producing department and houses, diet, drinking water samples, and technical ECF products were simultaneously collected for the analysis of PFAA isomers. The purposes of this study were to (1) study the temporal trends of the concentrations and profiles of PFAA isomers in occupational workers and the potential factors influencing the PFAAs profiles such as gender and the work assignment, (2) investigate the possible intake pathways of PFAA isomers in occupational workers, and (3) evaluate the daily clearance and a possible elimination mechanism of PFOS, PFOA, and PFHxS isomers. To our knowledge, this is the first systematic study to examine the intake, body burden, and excretion of PFAA isomers for workers involved in the manufacture of perfluorosulfonates.

■ MATERIALS AND METHODS

Chemicals and Reagents. Detailed nomenclature and structures, adapted from Benskin et al.,[29] are listed in Table S1 and Figure S1 in the Supporting Information. Standards of *n*-, 1*m*-, 3*m*-, 4*m*-, 5*m*-, *iso*-, (4,4)m_2-, (4,5)m_2-, and (5,5)m_2-PFOS; *n*-, 3*m*-, 4*m*-, 5*m*-, *iso*-, (4,4)m_2-, (4,5)m_2-, and *tb*-PFOA; an *n*-/*br*-PFHxS mixture; $^{13}C_4$PFOS; $^{13}C_4$PFOA; and $^{13}C_3$PFHxS were purchased from Wellington Laboratories (Canada). Methanol (HPLC-grade) was purchased from J.T. Baker (USA). Formic acid and ammonium hydroxide were purchased from Alfa Aesar (Ward Hill, MA, USA). Water was prepared using a Milli-Q Advantage A10 system (Millipore, USA). HLB (6 cc, 150 mg) and WAX (6 cc, 150 mg) cartridges were purchased from Waters Co. (Ireland).

Sample Collection. The PFSA manufacturing facility (Henxin Chemical Plant) is located in Hubei province, China, and is one of the largest PFOS-related producers in China. We were told that the production in the plant mainly involves the ECF process. Serum samples (n = 171) were collected from 36 volunteers from different departments, including the sulfonation department (SD), the electrolytic department (ED), the fabric-finishing-agent department (FD), the research building (RB), and the management office (MO) each November or December from 2008 to 2012. Urine samples were collected in 2011 and 2012 (n = 69). The serum and urine samples were all sampled in the morning, and the participants were told not to eat breakfast before the serum and urine sample collection. Detailed information about the workers is listed in Table S2. All volunteers gave their consent to participate in this study. A questionnaire was used to collect information about the department, work time, gender, dietary habits, age, weight, and height of the donors. After sampling, the serum was separated from the red blood cells and other components by centrifugation at 3000 rpm for 10 min. Then, the serum and urine samples were transferred as soon as possible to our laboratory in polypropylene containers and stored at −20 °C until analysis. The procedures were approved by the Ethic Committee of Research Center for Eco-Environmental Sciences and Medical Research Ethics Committee, School of Medicine, Jianghan University, and were in compliance with research requirements regarding human subjects.

Indoor dust samples and TSP samples were collected in 2011. In all, 28 indoor dust samples were collected, including 6 from different departments of the manufacturing facility (SD, FD, RB, MO, and 2 ED workshops), 9 from the workers' houses, and 13 from other residential housing around the facility. Additionally, 14 TSP samples were also collected, including 6 from different departments, 5 from the workers' houses, and 3 from other houses. Indoor dust samples were collected by sweeping the surfaces of furniture with precleaned brushes. The TSP samples were collected by a midvolume air sampler (Tianhong Intelligent Instrument Plant, Wuhan, China) with a Whatman quartz fiber filter (QFF). The flow rate was set at 120 L per minute for 24 h per sample.

Drinking water samples and duplicated diet samples were collected in 2012. Drinking water (tap water) samples (n = 2) were collected directly from the manufacturing plant. The workers have lunch in the canteen of the plant and have dinner in their own houses. Duplicate diet samples (rice = 9, dish = 8) were collected directly from the workers' dining tables in the canteen of the plant and the workers' houses.

Sample Preparation and Instrumental Analysis. Serum samples were extracted using an ion-pairing method. Indoor dust samples and TSP samples were extracted with methanol. Dietary samples were extracted with 10 mL of 50 mM KOH in methanol. The extraction and urine samples were then loaded onto HLB or WAX cartridges for further processing. Detailed information about the sample pretreatment is provided in the Supporting Information.

Analysis of the linear and branched PFAA isomers was performed using a HPLC-ESI-/MS/MS system, which consisted of a Waters 2695 Alliance high-performance liquid chromatograph and a Waters Quattro Premier XE triple-quadrupole mass spectrometer (Waters Corp., Milford, MA). Among the isomers, *n*-, 1*m*-, 3*m*-, 4*m*-, 5*m*-, *iso*-, and m_2-PFOS; *n*-, 3*m*-, 4*m*-, 5*m*-, *iso*-, and m_2-PFOA; and *n*- and *br*-PFHxS were detected. A method developed by Benskin et al. was adapted with minor modifications.[29] The final extract (10 μL) was injected onto a FluoroSep RP Octyl column (3 μ 100A, 15 cm × 2.1 mm, ES Industries). Methanol (A) and 5 mM ammonium formate (pH 4, B) were used as the mobile phases. The flow rate was set at 0.15 mL min^{-1}. The dual mobile-phase gradient started at 40% A; was held constant for 0.3 min; changed to 64% A by 1.9 min, 66% A by 5.9 min, 70% A by 7.9 min, 78% A by 40 min, and 100% A by 41 min; remained constant until 46 min; returned to the initial condition by 47 min; and then equilibrated for 13 min. The parent and product ions are listed in Table S1, and the chromatograms are shown in Figure S2.

Quality Assurance/Quality Control. The method limit of quantification (MLQ) was determined to be 10 times the signal-to-noise ratio in the actual samples. The individual MLQs are listed in Table S3. Matrix spike recoveries were carried out for all sample types in this study. The matrix spiked recovery ranged from 50.3 to 169%, and detailed information is shown in Table S4. One procedural blank was performed for every batch of seven samples. A PFAA standard of 2 ng mL^{-1} was used for quality control during the analysis. More detailed information about the quality assurance and quality control protocol can be found in the Supporting Information.

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Table 1. PFAA Isomer Concentrations in Serum, Urine, Indoor Dust, TSP, Diet and Drinking Water Samples

		PFOS							PFOA							PFHxS		
		n-	1*m*-	(3 + 5)*m*-	4*m*-	*iso*-	m_2-	∑	*n*-	3*m*-	4*m*-	5*m*-	*iso*-	*tb*-	∑	*n*-	*br*-	∑
serum (ng mL^{-1}, *n* = 171)	mean	2554	80.2	607	242	472	76.7	4032	993	61.5	3.29	6.67	25.6	nd	1090	1635	128	1763
	geomean	975	9.27	130	56.9	133	3.27	1386	284	0.24	0.22	0.30	1.14	nd	371	780	57.8	863
	median	922	13.0	157	63.2	129	10.1	1478	520	10.5	.nd	0.47	0.15	nd	537	938	66.0	995
	min	37.9	nd	nd	nd	2.20	nd	47.3	2.66	nd	nd	nd	nd	nd	2.66	11.1	1.06	12.8
	max	36625	1664	10700	4477	8429	1980	62898	10515	2571	180	401	1106	nd	14774	9799	1146	10546
	detected	171	150	166	167	171	113	171	171	97	80	89	87	0	168	171	171	171
urine (ng mL^{-1}, *n* = 69)	mean	1.17	0.01	0.21	0.10	0.36	0.09	1.94	2.73	0.09	0.22	0.15	0.23	0.01	3.43	4.28	1.49	5.77
	geomean	0.39	0.03	0.05	0.07	0.14	0.03	0.85	1.17	0.06	0.04	0.03	0.07	0.01	1.82	1.14	0.42	1.70
	median	0.46	nd	nd	nd	0.23	0.00	1.81	nd	nd	nd	nd	nd	2.11	0.46	nd	nd	nd
	min	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd	nd
	max	20.5	0.19	7.28	4.07	4.70	3.18	39.9	21.1	1.55	1.43	1.45	3.00	0.20	24.3	25.1	15.0	40.0
	detected	63	4	30	8	53	15	63	63	11	25	26	32	9	63	59	58	59
indoor dust (ng g^{-1}, *n* = 28)	mean	39941	315	5499	2009	3976	961	52701	4217	512	188	267	368	27.4	5579	14093	1633	15726
	geomean	626	0.64	53.2	16.1	40.5	0.62	830	307	3.42	3.71	8.47	10.4	0.73	360	13.3	2.24	18.6
	median	375	nd	26.9	11.3	34.2	nd	432	185	3.33	2.35	8.13	12.5	0.31	227	34.4	4.40	34.5
	min	nd	nd	nd	nd	nd	nd	nd	41.3	nd	nd	nd	nd	nd	41.3	nd	nd	nd
	max	489907	5274	71977	29230	48149	13807	658343	55518	13408	3882	5461	6122	750	85139	246335	24342	257201
	detected	27	9	26	23	24	14	27	28	20	25	27	24	15	28	21	15	21
TSP (ng/m^3, *n* = 14)	mean	5.28	0.00	0.77	0.31	0.49	0.13	5.28	62.47	8.49	5.33	6.74	9.77	0.99	93.78	0.88	0.10	0.98
	geomean	0.28	nd	0.03	0.01	0.04	0.01	0.40	0.70	0.05	0.07	0.04	0.05	0.01	0.94	0.08	0.01	0.09
	median	0.25	nd	0.03	0.01	0.03	nd	0.38	0.19	0.02	0.02	0.01	0.01	nd	0.26	0.09	0.01	0.09
	min	0.02	nd	nd	nd	nd	nd	0.03	0.03	nd	nd	nd	nd	nd	0.04	nd	nd	nd
	max	62.0	nd	9.19	3.63	5.16	1.61	78.0	994	103	63.1	82.5	121	12.3	1123	9.36	1.04	10.4
	detected	14	0	12	7	13	6	14	14	9	11	14	12	6	14	13	7	12
diet (ng g^{-1}, *n* = 17)	mean	2.48	0.08	0.40	0.21	0.43	0.10	3.56	0.94	0.03	0.03	0.06	0.04	nd	0.94	0.76	0.13	0.88
	geomean	0.69	0.06	0.07	0.08	0.09	0.05	0.83	0.88	0.02	0.02	0.04	0.03	nd	0.93	0.35	0.02	0.38
	median	0.56	nd	nd	nd	0.15	nd	0.72	0.95	nd	nd	nd	nd	nd	0.96	0.37	nd	0.37
	min	0.09	nd	nd	nd	nd	nd	0.09	0.42	nd	nd	nd	nd	nd	0.45	nd	nd	nd
	max	26.6	0.52	5.23	2.34	4.95	1.01	40.6	1.79	0.11	0.12	0.27	0.18	nd	2.47	4.22	1.12	5.34
	detected	17	1	8	2	13	2	17	17	2	2	3	1	0	17	15	5	15
drinking water (ng L^{-1}, *n* = 2)	mean	2.35	nd	0.39	nd	0.45	nd	3.19	2.10	nd	nd	nd	nd	nd	2.10	0.80	nd	0.80
	geomean	2.34	nd	0.39	nd	0.45	nd	3.19	2.09	nd	nd	nd	nd	nd	2.09	0.78	nd	0.78
	median	2.35	nd	0.39	nd	0.45	nd	3.19	2.10	nd	nd	nd	nd	nd	2.10	0.80	nd	0.80
	min	2.14	nd	0.38	nd	0.42	nd	2.96	1.88	nd	nd	nd	nd	nd	1.88	0.59	nd	0.59
	max	2.56	nd	0.40	nd	0.48	nd	3.42	2.32	nd	nd	nd	nd	nd	2.32	1.01	nd	1.01
	detected	2	0	2	0	2	0	2	2	0	0	0	0	0	2	2	0	2

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Daily Clearance Estimation and Daily Intake Calculation. The daily renal clearance of PFAAs was calculated on the basis of paired serum and urinary concentrations using eq 1.1, adopted by Zhang et al.[30]

$$CL_{renal} = C_{urine} \times V_{urine}/(C_{serum} \times W) \quad (1.1)$$

For females under 50,

$$CL_{total} = CL_{renal} + 0.029 \text{ mL day}^{-1} \text{ kg}^{-1} \quad (1.2)$$

where C_{urine} is the concentration of PFAAs in urine (ng L^{-1}), V_{urine} is the daily urine excretion volume ($V_{urine(female)}$ = 1.2 L day^{-1}; $V_{urine(male)}$ = 1.4 L day^{-1}), C_{serum} is the concentration of PFAAs in serum (ng mL^{-1}), and W is body weight (kg).

Estimated daily intake (EDI) of PFAAs for occupational workers via drinking water, diet, TSP, and indoor dust was calculated using the following equations.[24,28,31−33]

$$EDI_{diet} = C_{diet} \times M_{diet}/W \quad (2.1)$$

$$EDI_{DW} = C_{DW} \times V_{DW}/W \quad (2.2)$$

$$EDI_{TSP} = C_{TSP} \times V_{TSP} \times EF/W \quad (2.3)$$

$$EDI_{ID} = EDI_{ingestion} + EDI_{dermal\ absorption} = C_{ID} \times SIR \times EF/W + C_{ID} \times BSA \times SAS \times AF \times EF/1000/W \quad (2.4)$$

where C is the concentration of PFAAs. M_{diet} is the amount of diet, and V is the volume. The M_{diet} value of 542 g dw per day was based on a duplicate diet: V_{DW} = 2 L per day;[24] V_{TSP} = 28.8 m^3 day^{-1} (20 L min^{-1});[31,32] EF (exposure fraction) = 8/24 in the manufacturing facility, 12/24 at home, and 4/24 in other places on the basis of a questionnaire given to the participants; SIR (soil ingestion rate) = 0.05 g d^{-1}; BSA (body surface area) = 3692 cm^2; SAS (soil adhered to skin) = 0.096 mg/cm^2; AF (fraction of PFAAs adsorbed in the skin) = 0.03.[28,33]

Docking Analysis. The crystal structure of human serum albumin (HSA, code: 1H9Z) was extracted from Protein Data Bank and treated as the receptor. The protein atoms were typed using the CHARMM force field. The Flexible docking procedure was applied for docking. All the calculations were done with Discovery Studio 2.1 software. Detailed information on the docking analysis was described in the Supporting Information.

Statistical Analysis. Statistical analysis was performed using SPSS 17.0 software. PFAA concentrations below the MLQ were replaced with MLQ/2. Correlation was tested using Spearman's rank coefficients. A value of $p < 0.05$ was considered significant.

RESULTS AND DISCUSSION

Levels and Composition Profiles of PFAA Isomers in Serum and Urine Samples. Detailed information on the distribution of PFAA isomers in the serum of occupational workers is shown in Table 1 and Figure 1a. PFOS was the most abundant chemical among the three groups of PFAAs in the serum samples, with a geometric mean concentration of 1386 ng mL^{-1}. *n*-PFOS was detected in all serum samples (geometric mean concentration = 975 ng mL^{-1}) and was the predominant PFOS isomer, with a relative abundance between 37.9 and 97.3% (mean value of 63.3%) of ∑PFOS (the sum of PFOS isomers quantified). The geometric mean concentrations of the other PFOS isomers were ranked in the following order: *iso*-PFOS > (3 + 5)*m*-PFOS > 4*m*-PFOS > 1*m*-PFOS > ∑m_2-PFOS. Compared to results from previous studies on PFAAs in human blood samples (Table S5),[5,7,10,11,17,23,30,34,35] ∑PFOS in this study was very high, whereas the *n*-PFOS proportion was in a moderate range.

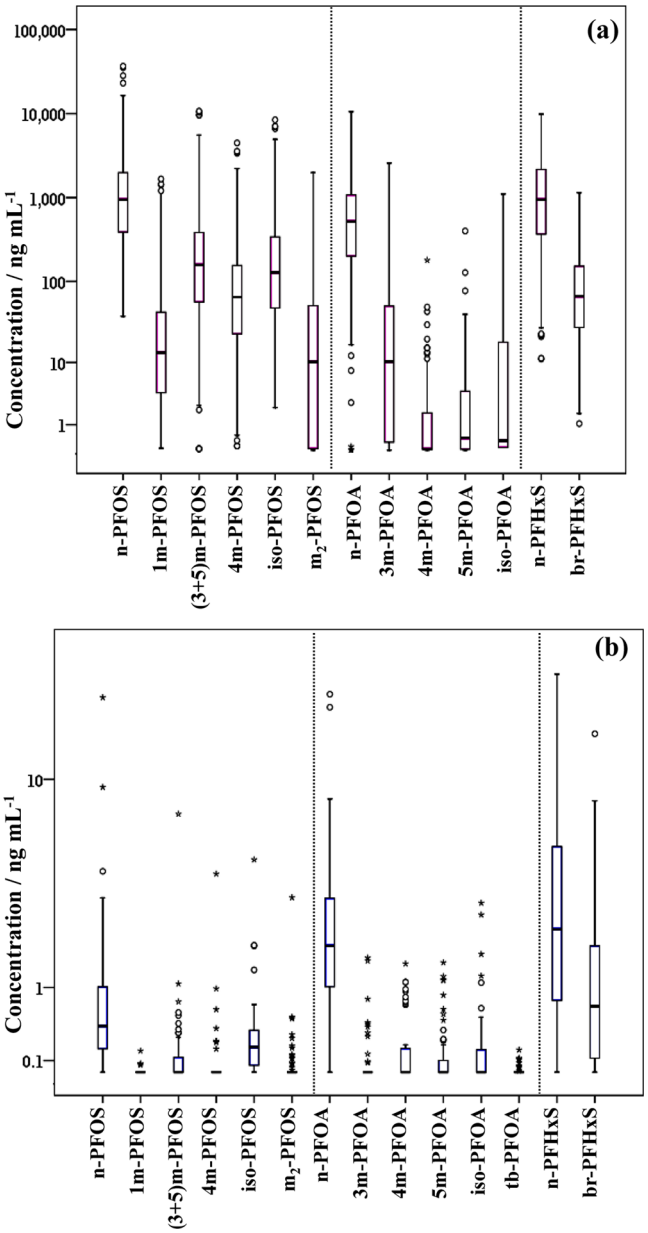

Figure 1. PFOS, PFHxS, and PFOA isomer concentrations in occupational workers' (a) serum samples and (b) urine samples. Boxes represent the 25th and 75th percentiles, three horizontal bars represent 5th, 50th, and 95th percentiles. °, outliers, and *, extreme values.

PFOA was detected in 171 serum samples of occupational workers, and the ∑PFOA (the sum of PFOA isomers quantified) ranged from 2.66 to 14774 ng mL^{-1} with a geometric mean concentration of 371 ng mL^{-1}. *n*-PFOA was the predominant isomer among the targeted PFOA isomers, and its concentration ranged from 2.66 to 10515 ng mL^{-1} with a geometric mean concentration of 284 ng mL^{-1}. The average proportion of *n*-PFOA was 91.7%, which was higher than that in the ECF products but lower than that observed in previous studies on the general population (Table S5).[5,7,11,17,30,35] For

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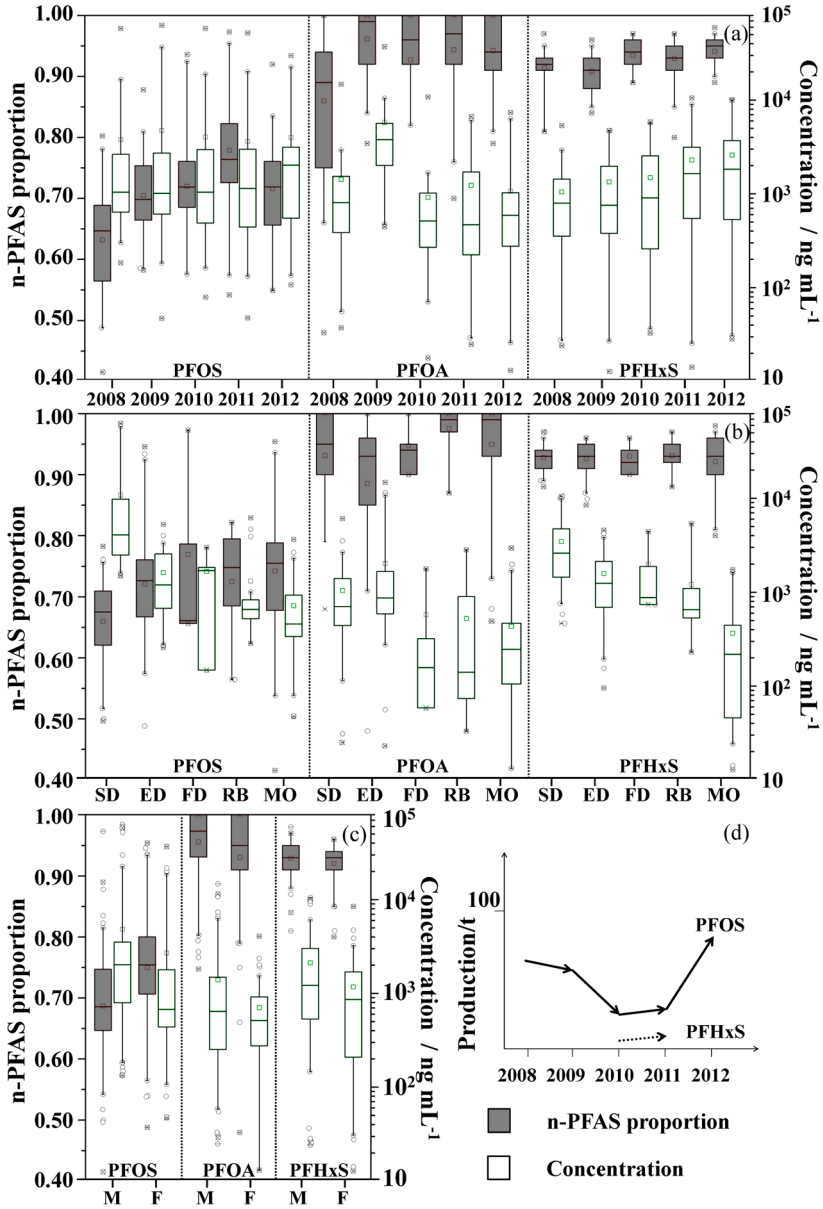

Figure 2. *n*-PFAA proportions and concentrations among occupational groups divided by (a) sampling time, (b) department (SD: sulfonation department; ED: electrolytic department; FD: finishing agent department; RB: research building; and MO: management office), and (c) gender (M: male; F: female). Boxes represent 25th and 75th percentiles, and three horizontal bars represent the 5th, 50th, and 95th percentiles.; °, outliers, and *, extreme values.

the individual branched PFOA isomers, the concentrations were ranked in the following order: *iso*-PFOA > 5*m*-PFOA > 3*m*-PFOA > 4*m*-PFOA.

PFHxS isomers have been found in carpet, dust, serum, and urine samples from a Canadian family with exceptionally high serum concentrations of PFHxS.[25] However, to our knowledge, quantification of PFHxS isomers has not yet been reported. In this study, the *n*-PFHxS and *br*-PFHxS concentrations were first separated and quantified in serum samples. ∑PFHxS (the sum of *n*- and *br*-PFHxS) was in the range of 12.8−10546 ng mL^{-1} with a geometric mean concentration of 863 ng mL^{-1}. The *n*-PFHxS proportion was 92.7% of ∑PFHxS. Overall, the PFHxS concentrations of the occupational workers were higher than those of populations under specific high exposure, such as the Canadian family who used Scotchgard and the fishery employees consuming contaminated fish from Tangxun Lake in China.[23,25]

The PFOS isomer concentrations in the serum samples were positively correlated with each other (Table S6), indicating that they shared the same source. Similar results were also found for the PFOA and PFHxS isomers. The relative abundance of *n*-PFOS was significantly negatively correlated with the ∑PFOS concentrations and individual PFOS isomer concentrations, and the individual branched PFOS isomer proportions were positively correlated with ∑PFOS, which was in accordance with other studies ($p < 0.05$) (Figure S3).[5,11] This implies that the branched PFOS isomers might be more prone to accumulate in serum with increasing ∑PFOS concentrations.

Figure 2a shows the temporal trend of the total concentrations of PFAA isomers and the *n*-PFAA proportions in the serum samples from occupational workers during the

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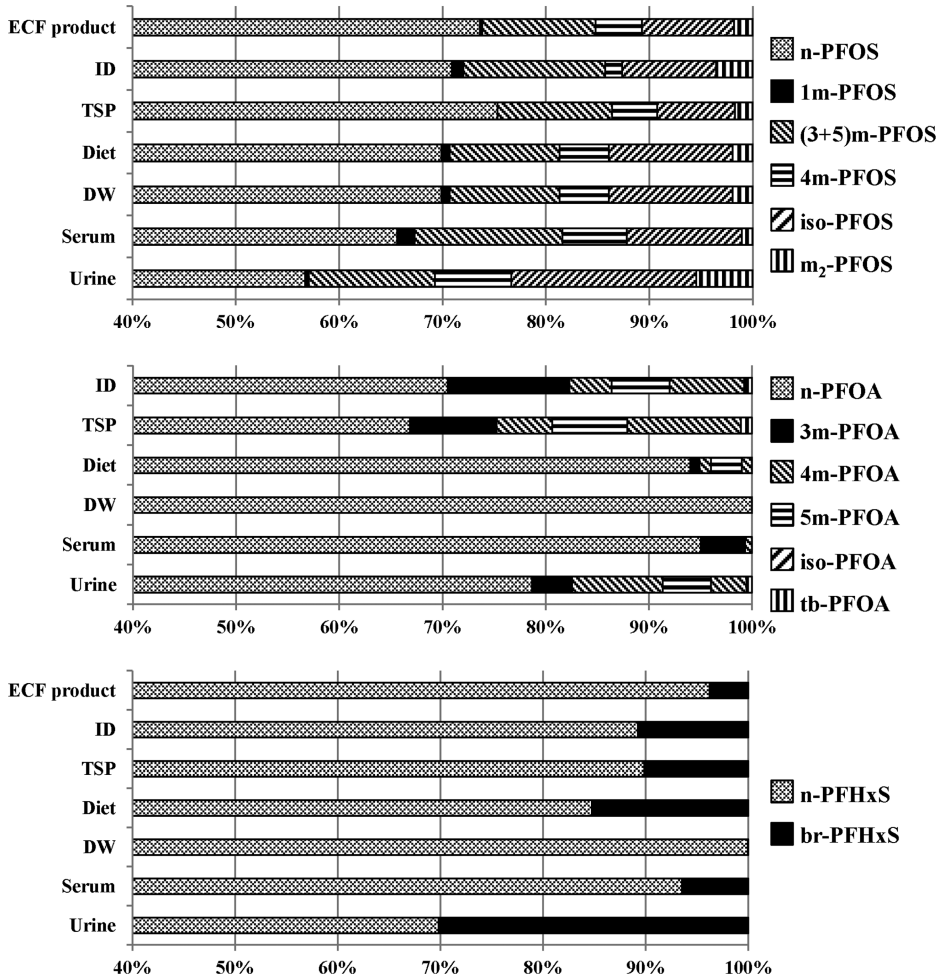

Figure 3. PFOS, PFOA, and PFHxS isomer profiles in the ECF product, indoor dust (ID), TSP, diet, drinking water (DW), serum, and urine samples of the manufacturing facility.

period of 2008−2012. The relative abundance of *n*-PFOS increased from 2008 to 2011 and decreased from 2011 to 2012 (Figure 2a), which was in contrast to the temporal trend of the ∑PFOS in serum and the annual PFOS production in this facility. The *n*-PFOS proportions were ranked in the following order: SD < ED < FD < RB < MO (Figure 2b). Additionally, the *n*-PFOS proportion was higher in females than in males (Figure 2c). The gender difference is believed to be related to specific excretion routes in females, including menstruation, placental transport, and breast milk.[36−38] The trends of the *n*-PFOS proportions were opposite to that of ∑PFOS when the variances among the sampling times, departments, and gender differences were taken into account, which corresponded to a negative correlation between the *n*-PFOS proportions and ∑PFOS ($p < 0.05$). The reason for this difference is unclear, but it was believed to be related to the different accumulation rate of PFOS isomers in humans. Generally, ∑PFOA and the *n*-PFOA proportion increased in 2009. The relative abundance of *n*-PFHxS increased slightly with increasing ∑PFHxS in the serum. However, if we considered the gender differences, we found that the *n*-PFOA and *n*-PFHxS proportions were constant in spite of the significant concentration differences between genders. The results further implied that the excretion rates of the PFAA isomers in males and females were different.

In urine samples, the detection rates of the PFOS, PFOA, and PFHxS isomers were 91.3, 91.3, and 85.5%, respectively. ∑PFOS, ∑PFOA, and ∑PFHxS were in the range of not detected (nd)−39.9, nd−24.3, and nd−40.0 ng mL^{-1}, respectively. The mean proportions of *n*-PFOS, *n*-PFOA, and *n*-PFHxS were 60.5, 79.8, and 74.1%, respectively. Generally, the proportions of the three linear PFAAs in urine samples were lower than those in corresponding serum samples from the occupational workers.

Correlation analysis indicated that *n*-PFOS was positively correlated with (3 + 5)*m*-, 4*m*-, and *iso*-PFOS in urine samples ($p < 0.05$, Table S7). For PFOA, *n*-PFOA was only correlated with 5*m*-PFOA and *iso*-PFOA ($p < 0.05$, Table S7), which might be due to the different renal excretion rates of PFAA isomers in humans. For PFHxS, the concentrations of *n*-PFHxS were significantly linearly correlated with *br*-PFHxS in urine samples ($R = 0.90$, $p < 0.05$).

Zhou et al. found that urine samples can be used as good matrices for biomonitoring the burden of PFASs in human bodies.[23] Li et al. found a positive correlation between PFOS in urine and serum samples but none for PFOA.[39] In this study, PFOS, PFOA, and PFHxS in the paired serum and urine samples showed significant linear correlations with each other (Table S8). For individual isomers, *n*-PFOS, (3 + 5)*m*-PFOS, *iso*-PFOS, *n*-PFOA, *iso*-PFOA, *n*-PFHxS, and *br*-PFHxS concentrations in the urine samples were significantly positively correlated with the corresponding concentrations in the serum samples ($p < 0.05$, Table S8). This result indicated that only

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these isomers in urine could represent the corresponding isomers in serum. Individual PFAA isomers behaved differently because of their potentially different transport mechanisms and elimination routes in humans. Besides, the low detection rates for 1*m*-, 4*m*-, and m_2-PFOS and 3*m*-PFOA isomers in urine samples may be a reason for the weak correlation between urine and serum samples.

Estimation of Intake of PFAAs via Different Routes of Exposure. Detailed information about PFAA levels in indoor dust is shown in Table 1. PFAA concentrations in indoor dust from the occupational settings and the workers' homes were much higher than samples collected from other places in this study and those from areas not affected by the production facility.[23,33,40] Generally, the PFOS, PFOA, and PFHxS isomer profiles in indoor dust from the worker's houses were similar to those from the facility, indicating that the PFAAs in the workers' houses could have originated from the manufacturing plant. In the TSP samples collected from the manufacturing facility, the geometric mean PFOS, PFOA, and PFHxS concentrations were 2.29, 24.1, and 0.69 ng/m^3, respectively. PFOA was inferred to be a byproduct of the electrolytic process because of the high PFOA concentrations in the indoor dust and the TSP samples from the electrolytic process department. The mean proportions of *n*-PFOS, *n*-PFOA, and *n*-PFHxS were 74.2, 66.9, and 89.9%, respectively. In the TSP samples collected from workers' houses, the geometric mean PFOS, PFOA, and PFHxS concentrations were 0.12, 0.09, and 0.02 ng/m^3, and the mean proportions of linear PFOS, PFOA, and PFHxS were 78.8, 79.5, and 91.7%, respectively. Overall, the proportion of *n*-PFOS and *n*-PFHxS in TSP from the work environment and the workers' houses were comparable to those in the technical products from this production facility (75.1% *n*-PFOS and 96.2% *n*-PFHxS isomer). In the other TSP samples, the mean PFOS, PFOA, and PFHxS were 0.29, 0.23, and 0.08 ng/m^3, and the mean linear isomer proportions were 78.1, 81.3, and 91.1%, respectively. Concentrations in TSP samples were positively correlated to indoor dust concentrations for PFOS and PFOA ($p < 0.05$), but not for PFHxS.

All of the target PFOS isomers were found in the dietary samples, whereas *tb*-PFOA was not detected. In meat and vegetables, the geometric mean concentrations of PFOS, PFOA, and PFHxS were 0.16, 0.23, and 0.04 ng g^{-1}, respectively. In the rice samples, the geometric mean concentrations of PFOS, PFOA, and PFHxS were 0.54, 0.80, and 0.13 ng g^{-1}, respectively. In the drinking water samples, the PFHxS concentration was 0.78 ng L^{-1}, and the PFOS concentration was 2.34 ng L^{-1}, with an *n*-PFOS proportion of 73.4%. The *n*-PFOA concentration was 2.09 ng L^{-1}. No branched PFHxS and PFOA isomers were detected.

In this study, indoor dust, TSP, diet, and drinking water were considered to be important direct routes for the intake of PFAAs. The PFAA isomer profiles in the technical products, the direct exposure routes (indoor dust, TSP, diet, and drinking water), and the serum and urine samples are shown in Figure 3. For PFOS, the *n*-PFOS proportion in the serum samples was lower than the intake, although *n*-PFOS in the urine samples showed lower proportions. However, *n*-PFOA and *n*-PFHxS were present at higher proportions in the serum samples than the intake, which could result from a faster renal clearance rate of branched isomers.

The average daily intakes of ∑PFOS, ∑PFOA, and ∑PFHxS for occupational workers via these four direct routes were 105, 57.5, and 32.5 ng d^{-1} kg^{-1}, respectively. Detailed calculations of the exposure/ingestion factors in eq 2 can be found in the Supporting Information. Overall, occupational exposure such as indoor dust intake, TSP intake, and diet were found to be the main exposure routes of PFAAs in the workers (Figure 4), which were rather different from those for the general populations.[18,22,24] For PFOS and PFHxS, the most predominant direct exposure pathway was via indoor dust intake, which accounted for 88.4 and 67.3%, respectively, followed by dietary intake, which accounted for 8.88% of PFOS and 31.6% of PFHxS. However, for PFOA, intake via TSP was the predominant route of exposure in the occupational workers, and it accounted for 67.9% of ∑PFOA, followed by indoor dust (17.2%) and diet (14.8%). TSP was more important for PFOA than for PFOS and PFHxS, especially in the electrolytic process department, where TSP accounted for 84.2% of the total daily intake of PFOA (Figure S4). To learn how much the work in the plant contributed, the contribution of indoor dust and TSP from the working place to the total intake was estimated. The proportions of the indoor dust intake in the working place to the total indoor dust intake were 99.6, 98.2, and 99.9% for PFOS, PFOA, and PFHxS individually. For the TSP intake, the proportions were 89.7, 99.0, and 93.2% respectively. Considering the huge variance of exposure distributions for PFOA in occupational workers in this study (Table S9 and Figure S4), the variance might stem from two factors. First, the formation or source of PFOA was different from that of PFOS/PFHxS because PFOA was the byproduct of the process. Second, the absorption of/adsorption to the particles of PFOA was different compared to that of the other two groups of PFAAs. Figure S5 shows that the ratio of TSP to indoor dust for PFOA from all of the five departments was

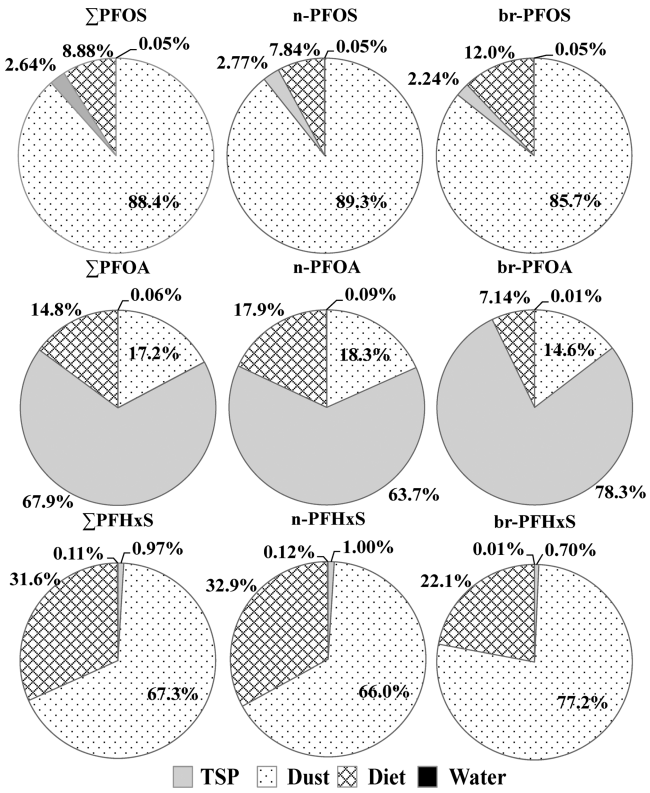

Figure 4. Estimation of PFAA intake of occupational workers via TSP, indoor dust, diet, and drinking water.

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Table 2. Renal Clearance of PFAAs Isomers (mL day^{-1} kg^{-1})

	PFOS							PFOA						PFHxS		
	n-	1*m*-	(3 + 5)*m*-	4*m*-	*iso*-	m_2-	∑	*n*-	3*m*-	4*m*-	5*m*-	*iso*-	∑	*n*-	*br*-	∑
geomean	0.01	0.11	0.02	0.03	0.04	0.10	0.01	0.09	0.07	1.19	0.38	0.19	0.10	0.04	0.18	0.05
mean	0.02	0.16	0.03	0.05	0.12	0.35	0.03	0.21	0.10	2.26	0.72	0.25	0.29	0.08	0.38	0.09
median	0.01	0.19	0.02	0.03	0.04	0.08	0.01	0.07	0.09	1.28	0.70	0.19	0.08	0.03	0.19	0.04
min	0.0002	0.01	0.001	0.01	0.004	0.005	0.0002	0.01	0.02	0.26	0.01	0.03	0.01	0.0003	0.01	0.004
max	0.07	0.25	0.18	0.13	1.24	1.31	0.20	2.17	0.23	13.6	1.81	0.64	6.53	1.19	7.55	1.43
n (valid)	61	4	28	7	52	12	61	61	8	13	12	23	61	57	56	57

higher than those for PFOS and PFHxS. Correlation analysis was conducted between PFAA concentrations in the serum and the estimated daily intake of the occupational workers among the five work departments. The results indicated that the PFOS and PFOA concentrations in the serum samples were positively correlated with estimated daily intake ($p < 0.05$), whereas they were not for PFHxS. However, there were also some limitations, such as the small sample sizes, especially for drinking water and diet samples.

PFAAs Clearance Rate and Elimination. Daily renal clearances (CL_{renal}) of the PFAAs were calculated on the basis of paired serum and urine concentrations (eq 1.1) and are shown in Table 2. The median renal clearances of ∑PFOS, ∑PFOA, and ∑PFHxS were individually 0.01, 0.08, and 0.04 mL day^{-1} kg^{-1}, respectively. Because of the low detection rate of the *tb*-PFOA isomer in serum samples, CL_{renal} of *tb*-PFOA isomer was not included. PFOA showed the highest renal clearance rate, followed by PFHxS and PFOS. Generally, CL_{renal} for branched isomers was higher than that for linear isomers. The geometric mean renal clearances of PFOS were ranked in the following order: 1*m*- → m_2- → *iso*- → 4*m*- → (3 + 5)*m*- → *n*-PFOS. For PFOA, the CL_{renal} values were ranked as follows: 4*m*- → 5*m*- → *iso*- → *n*- → 3*m*-PFOA. Overall, the CL_{renal} values in this study were higher than the results of Zhang et al. for the general population.[30]

Upon comparing the daily intake and renal clearance rates, we found that elimination of the PFAAs via excretion in the urine only accounted for a very small part of the entire elimination process in occupational workers (Table S10). Apart from renal excretion and menstrual clearance, feces, sweat, and breast milk are also important excretion routes for PFAAs.[25,37−39,41,42] Previous studies showed that elimination through the feces or reabsorption by the intestinal tract might play a more important role for PFOS than for PFOA.[43,44] Further investigation of the importance of the other elimination routes for this study group is warranted.

HSA, the most abundant protein in human blood, is a multifunctional carrier protein, and it can be found in the interstitial fluid of body tissue.[45] Previous studies on the binding of PFAAs to HSA have found that PFOS could be transported by binding to HSA.[45−47] To study further the potential differences in the transport behavior of different PFAA isomers in humans, the binding mechanisms between PFAA isomers and HSA were constructed on the basis of a docking approach, and the interaction between PFAA isomers and HSA was indicated by the Rerank Score (Table S11 and Figures S6 and S7). A lower Rerank Score indicates a stronger interaction. Clearly, *n*-PFOS has a different docking mechanism from the branched isomers. For example, *n*-PFOS was inserted deeply into the binding pocket. In addition, several ionic and polar residues involved in the electrostatic interaction or hydrogen-bonding interaction were in proximity to *n*-PFOS, which also has an important function in stabilizing the PFOS−HSA complex via electrostatic interaction as well as via hydrogen bonds formed between the Ser427 residue and the O atom of the *n*-PFOS compound. Hydrogen-bonding or electrostatic interaction functions as an anchor, which determines the 3D spatial orientation of the *n*-PFOS compound in the binding pocket. A different mode was observed for *iso*-PFOS compounds, which showed quite a different orientation in the HSA binding site. Our results also revealed the diverse residues that were involved in the electrostatic interaction compared with *n*-PFOS. We also found that no hydrogen bond formed between HSA and *iso*-PFOS, which was in agreement with the docking score of the two compounds (Table S12). The results generally showed that the interaction between *n*-PFOS and HSA was stronger than that for the branched isomers, which indicates that *n*-PFOS has a greater potential to be transported to other tissues through binding with HSA, which would further decrease the proportion of *n*-PFOS in human blood. For PFOA and PFHxS, the linear isomers also showed stronger interaction with HSA than the corresponding branched isomers, although the proportions of linear PFOA and PFHxS isomers in serum samples were different from that of *n*-PFOS, which has also been confirmed by the work of Beesoon et al.,[48] so the stronger binding affinity of linear PFAAs may result in urine excretion rates of linear PFAAs isomers that are lower than those of their corresponding branched isomers.

Dust and TSP intake were the main exposure pathways of PFAAs for the occupational workers in this plant, which implied that appropriate occupational protection can help the workers to decrease the risk levels of PFAAs exposure. The ratio of *n*-PFOS decreased with the increasing concentrations of PFOS in human blood. It was presumed that the interaction with HSA and the difference in renal excretion and other excretion routes would jointly result in the different PFAA isomer profiles in the blood of humans compared to the compositions of the PFAAs in production. High concentrations of PFOS, PFOA, and PFHxS implied that the health effect on occupational workers caused by PFOS and its potential alternatives such as PFHxS should still warrant appropriate occupational protection under the working conditions of ongoing high exposure to PFAAs.

■ ASSOCIATED CONTENT

Supporting Information

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Notes

The authors declare no competing financial interest.

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